Data Science 6

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Lecture 10 *Monday, June 30, 2026*

Condtionals and something else I coudn't remember

0.1 Booleans

Recall Booleans and comparisons. Python reads from left to right. Today we learn 3 new general comparison ops: **and**, **or**, and **not**. XOR (exclusive or) only one can be true.

Note. Quiz question: Truth table

We can also make custom boolean preds instead of using **are**:

1. Define a function that returns a boolean value based on row values
2. **apply** that creates a new function of boolean values
3. use **where** to filter rows based on new boolean column
4. Drop the boolean column and have original table

Nah bruv just use a lamda function instad with .where and .apply .

*Note. Booleans can short circuit, this is where **and** for example will see a first false value and return true and for or once it finds a true value will short circuit and return a true value. **and** returns last true value*

Weird example: "hello" or (1 / 0) evaluates to "hello", because python will return the first short circuited value. So for example **or** it will return first truthy value which is why "hello" is returned.

🔍 **Practice and** and **or** short circuit

0.2 Conditionals

Use **if** statments using boolean condtionals. Recall **elif** and **else**.

Note. IN IF STATMENTS YOU ARE IN GLOBAL SCOPE

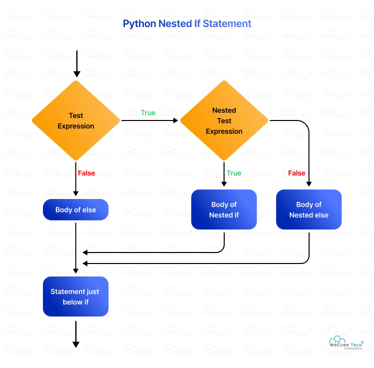

Figure 1: If statment diagram

Recall indexing in lists. You can use diffrent indexing tricks like using -1 to grab the last element or **slicing** lists or text or arrays (basically any "colleciton"). To remeber slicing its basically the same notation as **np.arange()**.

Another trick you can use is Containment using **in** to check if something is contained in another.

0.3 Random Other Stuff